Time series project: Spain Exports

Data Analysis

Degree in data science and engineering

UPC

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1. Introduction

This document is the report of the study of the *export* time series, which contains data of the total exports of Spain from 1999 to 2018. The main objective has been to find a proper seasonal ARIMA which fits the data in order to be able to make good and reliable predictions.

The necessary mathematical steps have been automatized using r, and this study includes the code used to read and transform the time series, find some possible models, evaluate them, make predictions and choose the best one. Not everything is included in this report, only the most important information necessary to understand how the study has been developed. In order to check details, please open the other file with r studio or another suitable interface.

2. Dataset description

The dataset contains the value of the total exports of Spain in billions of euros, given as a monthly value, dating from January 1999 to December 2018. The values are of the current gross price (price at the moment of the transaction, without inflation nor taxes applied).

The data was originally given by the Industry and Tourism Ministry of the Spain Government, in a web site known as Badase[1]. That link seems to not be working at the moment of writing this report, so we have used the datafile proportioned by the teacher of this subject, included in the Series Repository of Data Analysis.

The following graphic shows the data plot.

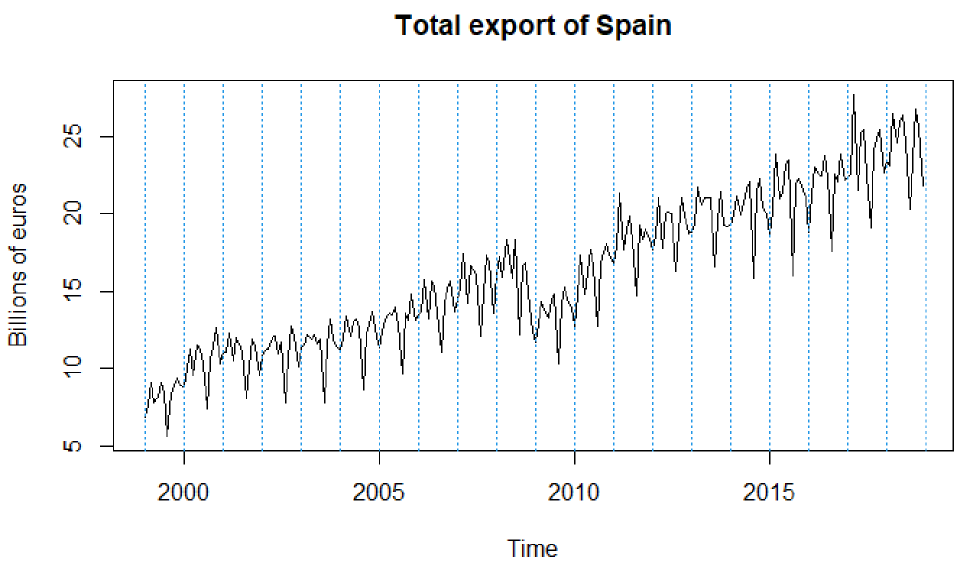

Figure 1 Original time series X_t: Exports of Spain in Billions of euros

By doing an initial visual analysis, we can see clearly see an increasing long time average, as well as a pattern for the short term average: each year shows a spike around summer, showing a sudden decrease in export value. We can also notice some outliers, one of them being extremely important to consider: the values right after the Great Recession (economic crash of 2008).

3. Used methodology

We have followed the Box-Jenkins methodology, a common statistical approach for time series analysis and forecasting. It consists on identifying, estimating and evaluating some possible seasonal ARIMA models before using them to predict future data.

First of all, the data has been read and plotted, and an initial visual evaluation has allowed to make the initial transformations in order to make the series stationary: Box-Jenkins transform and seasonal and regular difference appliance.

With the transformed series, we have searched for possible seasonal ARIMA models using the ACF and PACF plots, erased unnecessary parameters and observed the AIC to be able to select some good models.

We have evaluated normality, independence of residuals and homoscedasticity of residuals of the selected models, fitted them excluding the last year of observations and then predicted these excluded observations, also checking the stability.

Lastly, outliers have been treated and calendar effects have been mitigated, and predictions for 2019 have been made.

The R code file contains all the code used, with comments made for each step, and the convenient plots. We will only use in this report the most important ones.

4. Results and interpretation

a) Identification

We have considered that the best transformed series would be the one with the logarithm applied, along with one seasonal difference and one regular difference. This gives a constant variance and mean very close to 0, thus being a stationary series.

The seasonal difference is necessary to eliminate the seasonal component, which appears clearly when making the monthplot, as we can observe that the means for each month are different.

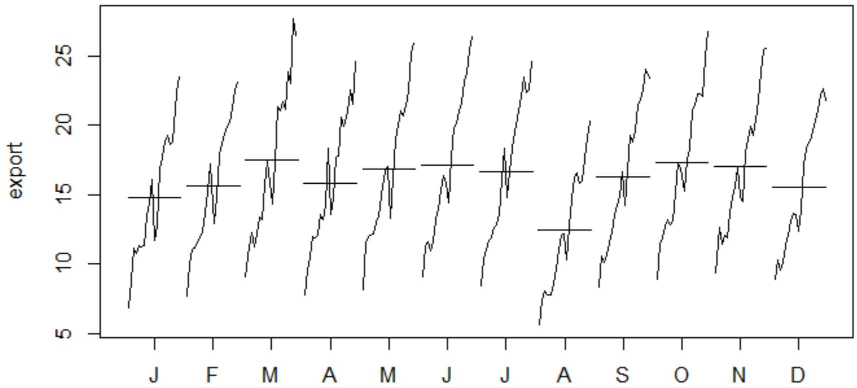

Figure 2 Monthplot of the original export series

After applying the seasonal difference, we see a series which does not seem to have a constant mean, so we should apply a regular difference. After this our series has a near 0 mean value and has reduced its variance.

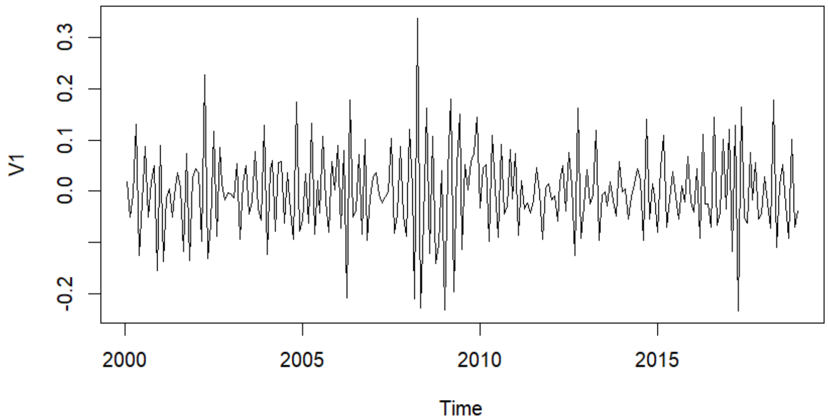

Figure 3 Stationary series ($W_t = (1-B)(1-B)^{12} \cdot log(X_t)$)

Applying another regular difference, increases the variance of the series; thus, this differentiation is rejected.

With the series properly transformed, we have plotted the ACF and PACF in order to find possible seasonal ARIMA models (made from AR and MA components).

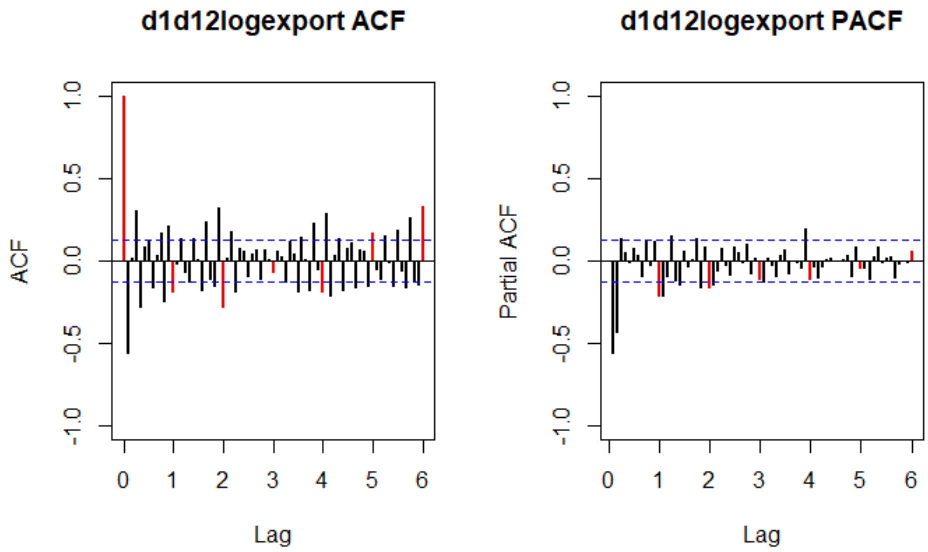

Figure 4 ACF and PACF of the stationary series

By looking at the graphics, it seems clear to us that there must be an infinite number of non-zero lags at the ACF for the regular and for the seasonal component. This implies that the suitable models for both components will be AR models.

For the seasonal component, it is reasonable to consider that lags 12, 24, 36 and 48 are different from zero, because the 36th and 48th can have fallen inside the confidence band by chance. It is also fair to consider that one or both are not statistically different from zero.

For the regular component, we clearly see that the first two are different from zero, but the 3rd could be statistically zero and have fallen outside the confidence band by chance.

Because the number of models can be quite high for some of them, we have tried ARMA alternatives with the same or less number of parameters. As there are a high variety of possible models, we only have considered some of them. This is the list of the most important models in the common SARIMA(p, d, q)(P, D, Q)$_s$ format.

- SARIMA(3, 1, 0)(2, 1, 0)$_{12}$
- SARIMA(2, 1, 0)(2, 1, 0)$_{12}$
- SARIMA(2, 1, 0)(4, 1, 0)$_{12}$
- SARIMA(2, 1, 0)(2, 1, 2)$_{12}$

b) Estimation

R code allows to see the AIC for any model easily, which helps to make some first decisions on studying or not similar models. We have taken advantage from this and seen that the 3rd and 4th of the models listed before, seem to be able to explain more variability of the data than the 1st and the 2nd, so those will be the models to focus on. Moreover, all model coefficients are clearly significant using t-statistic criterion.

- SARIMA$(2, 1, 0)(4, 1, 0)_{12}$:

$(1 + 0.7124B + 0.3647B^2)(1 + 0.5446B^{12} + 0.6144B^{24} + 0.4264B^{36} + 0.4153B^{48}) (1 - B)(1 - B^{12}) \ln(X_t) = Z_t$

- SARIMA$(2, 1, 0)(2, 1, 2)_{12}$:

$(1 + 0.736B + 0.387B^2)(1 - 0.7662B^{12} + 0.6129B^{24})(1 - B)(1 - B^{12}) \ln(X_t) =$
$= (1 + 1.3540B^{12} - 0.7535B^{24}) Z_t$

c) Validation

We will do the complete process with each model for separate.

1. Model 1: SARIMA$(2, 1, 0)(4, 1, 0)_{12}$

By checking AR and MA polynomials' roots modules, we see that all greater than 1 in absolute value, meaning the model is both invertible and causal. This means that the model can be expressed both as an infinite AR and MA.

We will proceed with the residuals analysis to check normality, independence and homoscedasticity. First of all, we will check the residuals plot and the square root of the absolute value of the residuals, to try to validate homoscedasticity.

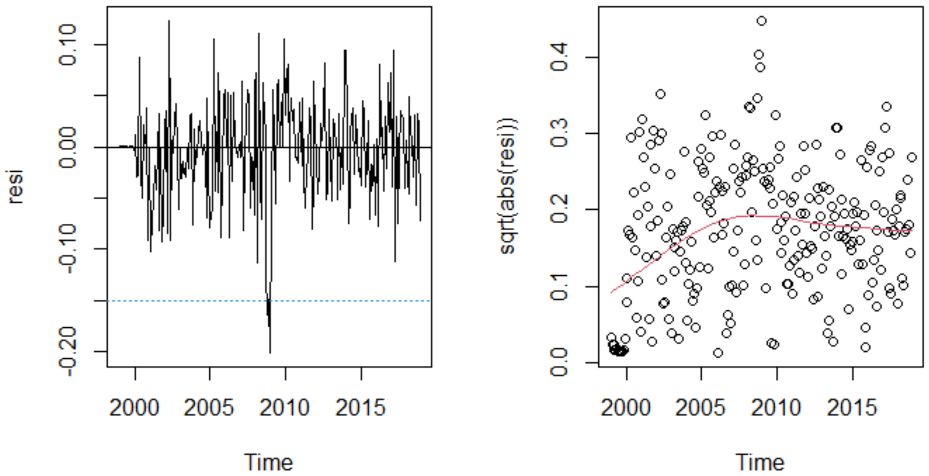

Figure 5 Residuals (left) and square root of residual absolute value (right) of SARIMA(2, 1, 0)(4, 1, 0)$_{12}$

They allow us to check the behaviour. The fact that the first plot does not show any weird pattern and that most of the residuals inside the confidence band tells us that the variance is constant enough. Nevertheless, we notice an outlier before 2010. We will take it into account for later. The second's plot line is not horizontal, which a sign that variance might be not so constant, as it seems to be lower for the first observations. We can consider that the model does not have constant variance, but that does not mean we can't work with it and use it to make predictions.

We will proceed to check normality using the Q-Q plot and the Shapiro-Wilks test.

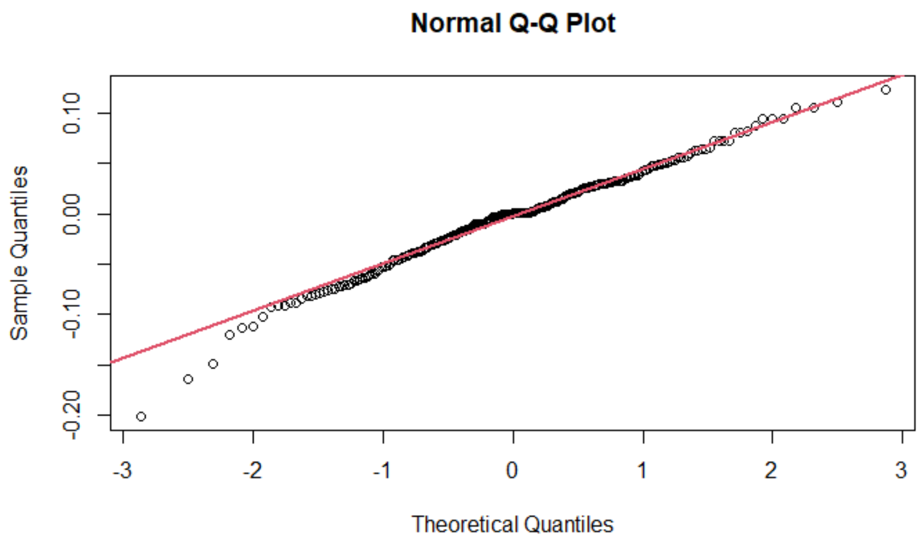

Figure Normal Q-Q plot of the residuals of the of (2, 1, 0)(4, 1, 0)$_{12}$

We can see that the lower tail gets a bit deviated from the red line, but it's not a deviation strong enough to consider that the residuals do not follow normality.

The Shapiro-Wilks test results in a p-value of 0.019 lower than 0.05. Thus, we could statistically refuse normality on the residuals, however considering normal Q-Q plot this is probably caused by outliers.

Lastly, we will use the plot of the ACF / PACF of the residuals and the Ljung-Box test to check independency.

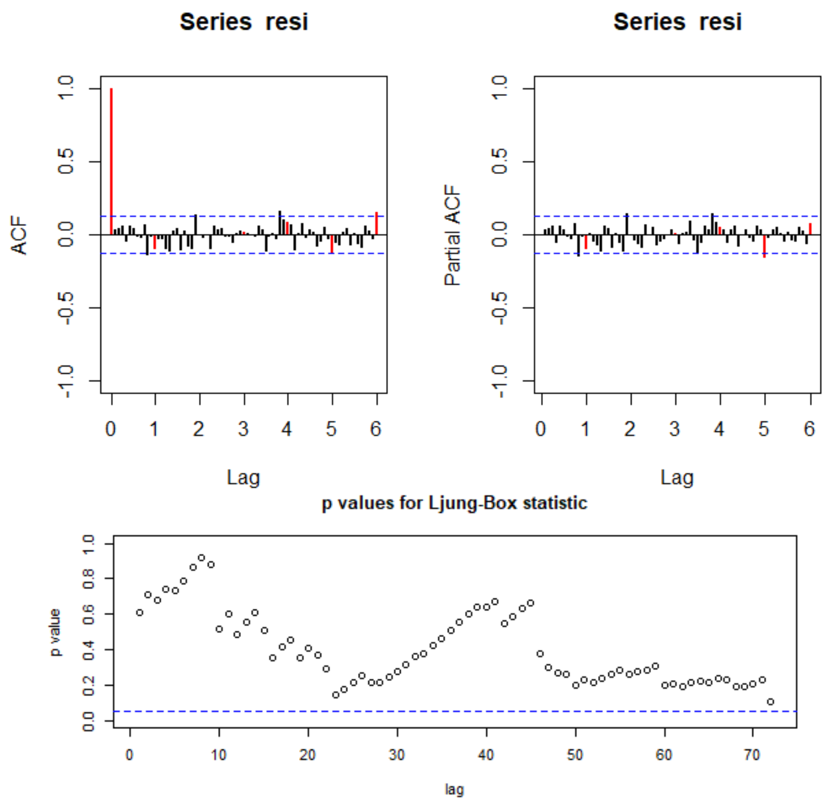

Figure 6 ACF, PACF and Ljung-Box statistic of the residuals of the SARIMA(2, 1, 0)(4, 1, 2)$_{12}$ model

We can see that most of the lags fall inside the confidence band, and the ones that do not only surpass it by a little, which could be by chance. By looking at the p values, we see that all of them are over the probability band, which is a very good indicator that residuals can be considered to be independent.

To sum up, we have properly validated normality and independence of residuals, but variance does not seem constant enough.

Now we will take a look at some common model fit metrics.

- AIC: -673.88
- BIC: -649.90

The accuracy of this model can be checked by fitting it excluding the last 12 observations and predicting them. It is necessary to check that the model is stable. We see that estimated parameters and their standard deviation are almost the same remaining with the same sign and significance, although the AIC Is lower. However, this makes sense because the model is based on data which is now missing. The model is stable.

The following plot shows in black the original data, in red the predicted values, and in blue the confidence bound (so we can assume that, from 100 predicted values, 5 will have their real data outside this interval).

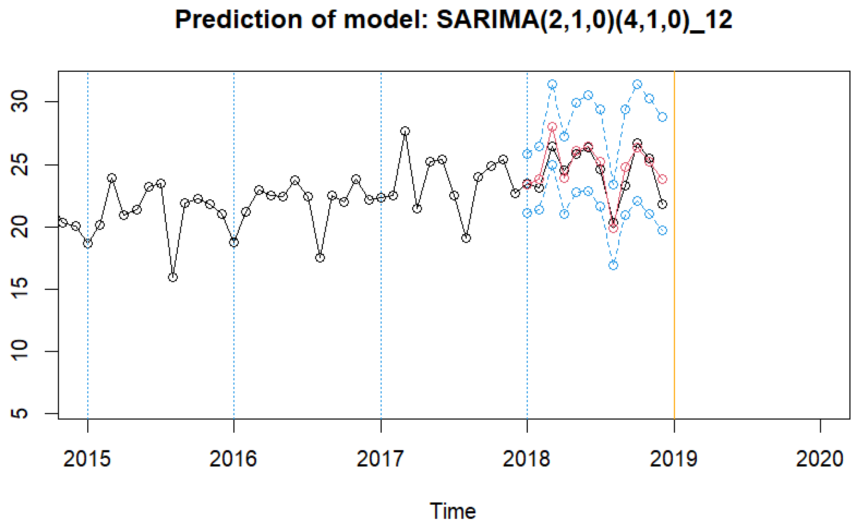

Figure 7 Prediction for 2018 of the model SARIMA(2, 1, 0)(4, 1, 0)$_{12}$

The prediction appears to be accurate. We should consider some metrics:

- RMSE: 0.917
- MAE: 0.699
- RMSPE: 0.039
- MAPE: 0.0294
- Mean confidence interval amplitude: 7.332

The mean absolute error is quite different from the root mean square error, meaning that predictions could vary more than the actual 5% of confidence. It does not mean that predictions won't be accurate, but it is something to take into account.

2. Model 1: SARIMA(2, 1, 0)(2, 1, 2)$_{12}$

The same process has been made with this model. First of all, like the previous model, all its polynomials roots have a modulus greater than 1 in absolute value. By consequence, this model is both invertible and causal, and can be expressed as an infinite MA and AR.

Checking the residuals plot and the square root of the absolute values, we see almost the same behaviour as the other model does:

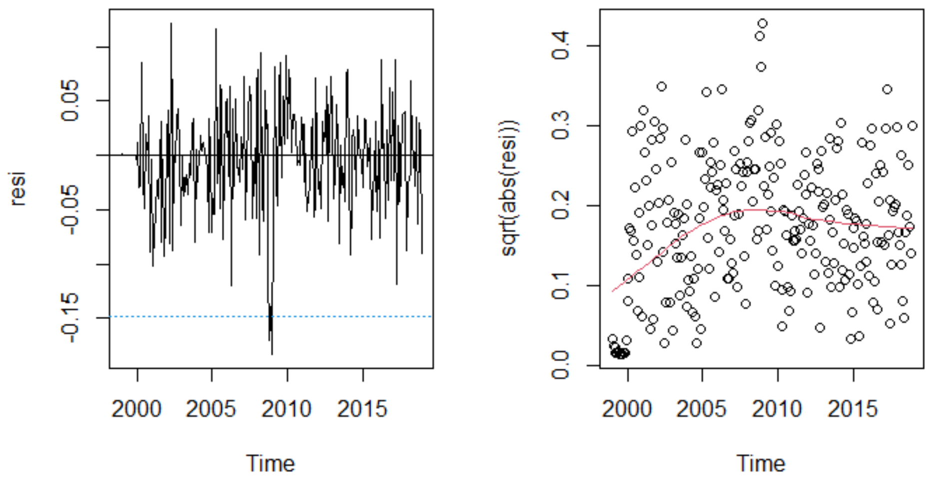

Figure 8 Residuals (left) and square root of residual absolute value (right) of SARIMA(2, 1, 0)(2, 1, 2)$_{12}$

So we can consider that variance is neither constant for this model. Normality has been checked the same way, using the Q-Q plot and the Shapiro-Wilks test.

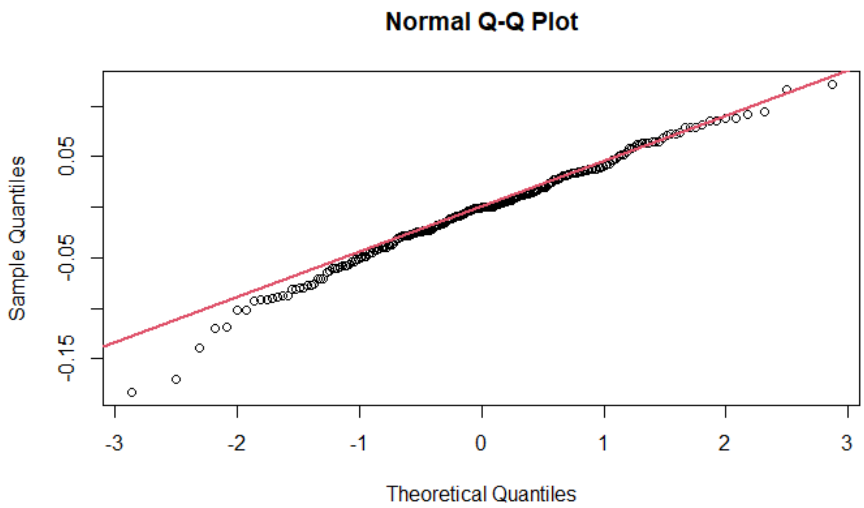

Figure 9 Normal Q-Q plot of the residuals of the of SARIMA(2, 1, 0)(2, 1, 2)$_{12}$ model

In this case, the points at the tails have higher deviation, and it is reasonable to consider that this model does not follow normality as good as the first.

For this model, the Shapiro-Wilks test results in a p-value of 0.054 higher than 0.05. Thus, we can't statistically refuse normality on the residuals.

Lastly, independence has been checked using the ACF / PACF of the residuals and the Ljung-Box test

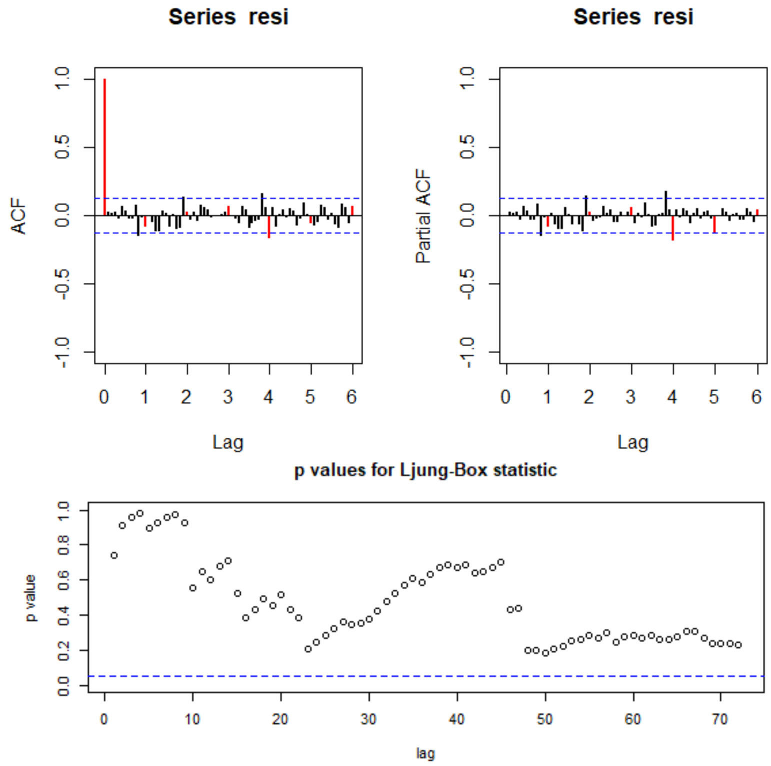

Figure 10 ACF, PACF and Ljung-Box statistic of the residuals of the SARIMA(2, 1, 0)(2, 1, 2)$_{12}$ model

The behaviour is similar to the first model's: most lags are inside the confidence band at the ACF and PACF, and all the p values for the test are high enough, meaning that we can consider this model to have independent residuals.

To sum up, this model is pretty similar to the other one (not constant variance, independent residuals), but normality does not seem to be achieved.

These are the same model fits metrics in this case:

- AIC: -674.87
- BIC: -650.90

Again, we will fit the model excluding the last year's data and try to predict it. Estimated parameters and standard deviations are also very similar so this model also seems to be stable.

This is the resulting plot:

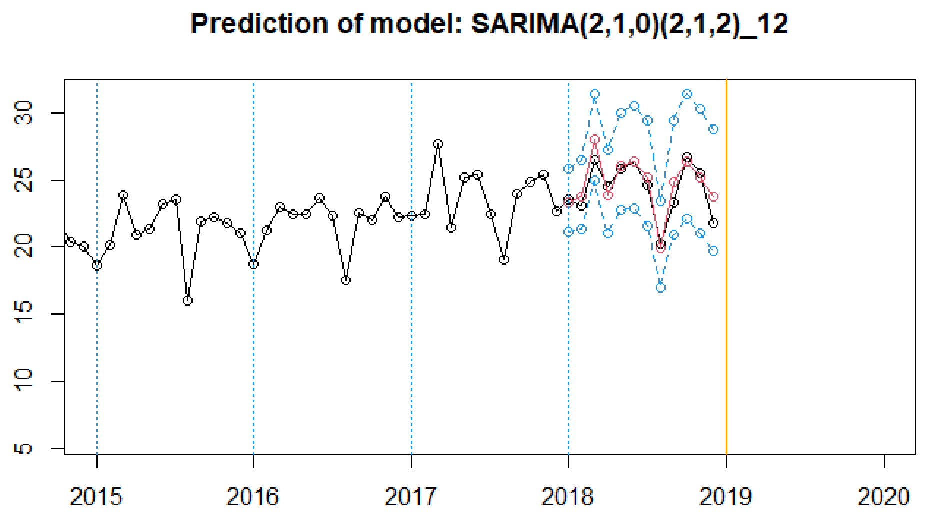

Figure 11 Prediction for 2018 of the model SARIMA(2, 1, 0)(4, 1, 0)$_{12}$

This prediction also appears to be accurate. Let us look at the same metrics:

- RMSE: 0.973
- MAE: 0.757
- RMSPE: 0.043
- MAPE: 0.032
- Mean confidence interval amplitude: 7.120

Again, the root mean squared errors are a bit higher than the absolute errors, due to there being a great number of "big errors" compared to "littler errors".

3. Choosing the best model for forecasting

The first model seems to behave better with respect to normality and has a little lower RMSE, MAE and mean confidence interval amplitude. Although they are not very different, we will focus on the first one: SARIMA(2, 1, 0)(4, 1, 0)$_{12}$, which has better properties for forecasting.

d) Forecasting

This is the plot of the predictions for 2019 using the chosen model (SARIMA(2, 1, 0)(4, 1, 0)$_{12}$):

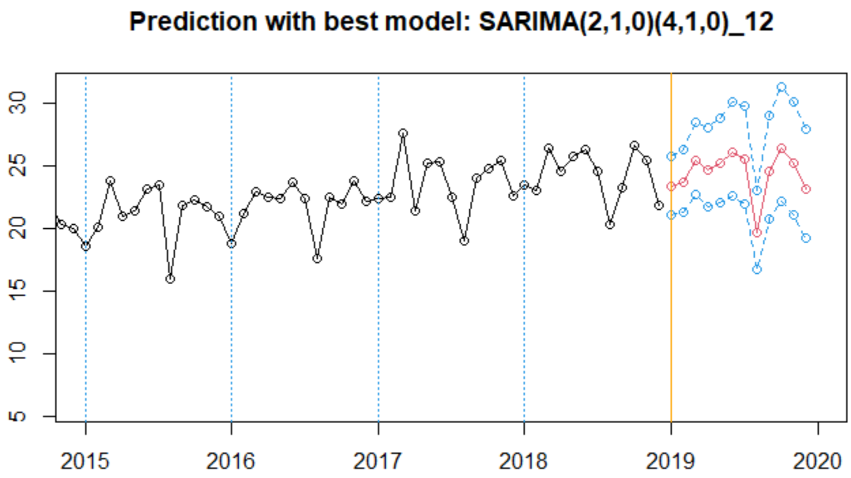

Figure 12 Predictions of the 12 months passed last observation with model SARIMA(2, 1, 0)(4, 1, 0)$_{12}$

e) Calendar effects

Because the data of this series is related to economics, calendar effects should be considered. The chosen model has been fitted removing calendar effects using the file CalendarEffects.r, proportioned by the teacher. The resulting model behaves very similarly to the one without the effects considered, but independence of residuals seems much clearer now:

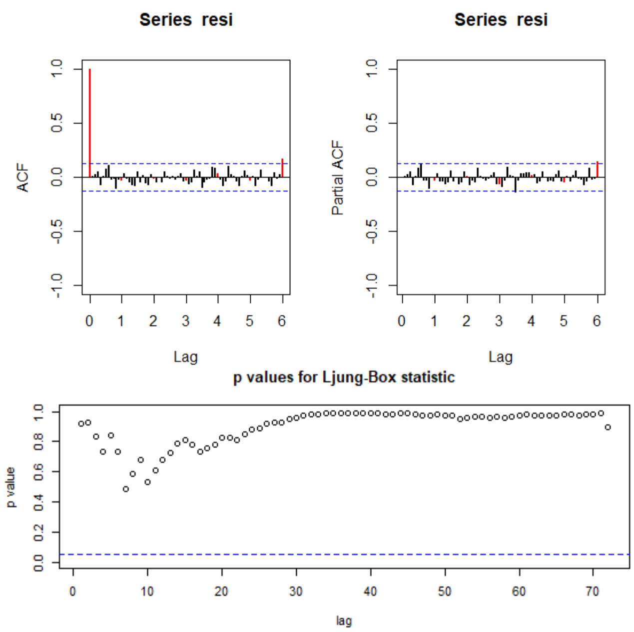

Figure 13 ACF, PACF and Ljung-Box test for residuals of AR model with calendar effect coefficients

T the residuals ACF / PACF show less lags outside the band, and p values for the Ljung-Box test are much higher, thus independence is clear for this model.

It's necessary to find the new model with the found coefficients, by analyzing the ACF / PACF.

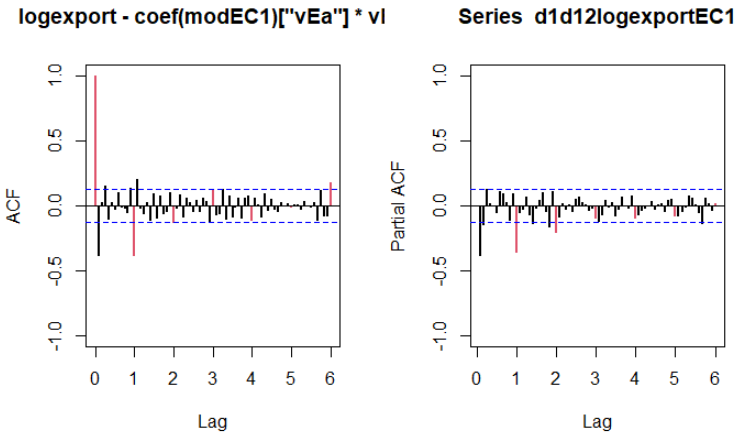

Figure 14 ACF and PACF of series adjusted for calendar effects

We have found three possible models, but chose a SARIMA(2, 1, 0)(0, 1, 1)$_{12}$ from between them.

Now we should compare the previous SARIMA(2, 1, 0)(4, 1, 0)$_{12}$ with this found model. The last 12 observations have been predicted with the model fitted excluding them. They have similar predictions, but metrics show the better model.

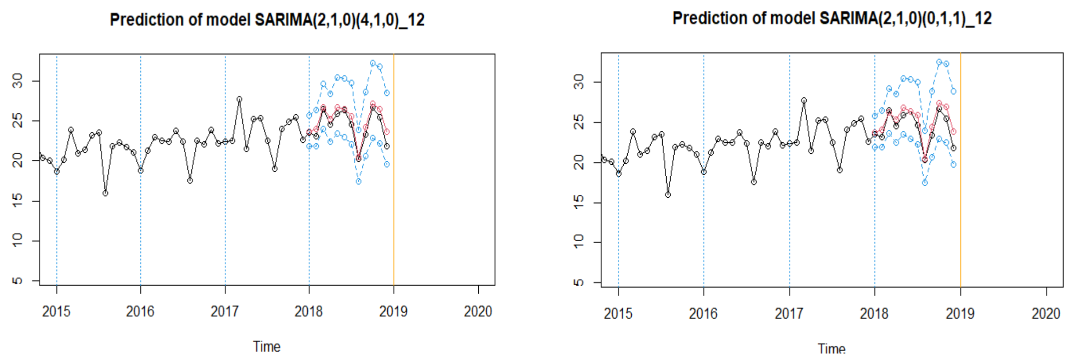

Figure 15 Predictions for models with calendar effects: SARIMA(2,1,0)(4,1,0) (left), SARIMA(2,1,0)(0,1,1) (right)

Metric	SARIMA(2, 1, 0)(4, 1, 0)$_{12}$	SARIMA(2, 1, 0)(0, 1, 1)$_{12}$
RSME	0.834	1.006
MAE	0.674	0.817
RMSPE	0.036	0.043
MAPE	0.028	0.034
Mean conf. int. amplitude	7.008	7.100

Table 1 Prediction metrics of the 2 calendar models

This new model is not better for predicting than the previous AR model with calendar effects applied, so because our main goal is to make predictions, we are keeping the SARIMA(2, 1, 0)(4, 1, 0)$_{12}$ considering calendar effects.

f) Outlier treatment

While validating the former models we have detected some unexpected values at the tails of the Q-Q normality plots. This can be caused by the presence of outliers in our data.

We can study the presence of outliers with automatic outlier detection over the best model we have so far: SARIMA(2, 1, 0)(4, 1, 0)$_{12}$ considering calendar effects. Using the default parameters from the atipics2.R file, given by the teacher, we obtain the following outliers:

Obs	Type	Fecha	W_coeff	Perv
23	TC	Nov 2000	0.08803784	109.20294
108	AO	Dic 2007	0.08233027	92.09677
114	AO	Jun 2008	0.09069734	91.32941
118	LS	Oct 2008	0.08691426	91.67557
120	LS	Dic 2008	0.20194757	81.71378
136	AO	Abr 2010	0.08273382	92.05962
171	AO	Mar 2013	0.09165300	109.59844

Table 2 Outliers detected on the logseries data

Analysing the dates of the identified outliers it is hard to find a specific reason for the one on November 2000. However, the outliers between 2007 and 2008 can be linked to the global financial crisis. The one in 2010 can be also related to the slow recovery from the crisis, while the 2013 outlier can be sign of economic stabilization after the crisis.

We can create a series without the effect of outliers. If we subtract the outlier-adjusted series from the original, we can observe the effect of the outliers.

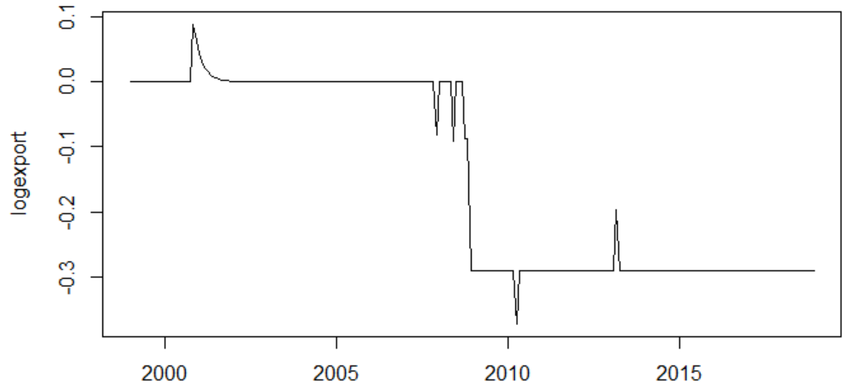

Figure 16 Plot of log(exports) – log(exports) (outlier adjusted)

We can now use the adjusted series to fit a new model using the new series ACF and PACF

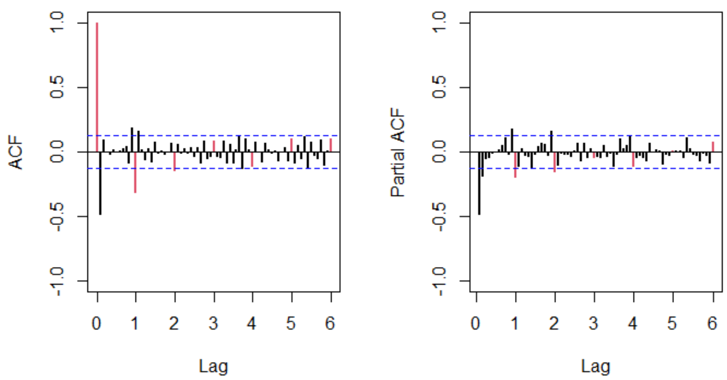

Figure 17 ACF and PACF of the outlier adjusted series

After analysing the ACF and PACF and trying some models we chose to study SARIMA(2, 1, 0)(4, 1, 0)$_{12}$ and SARIMA(0, 1, 1)(4, 1, 0)$_{12}$, both considering calendar effects since it have been proved meaningful.

Now we must compare both model predictions metrics. The last 12 observations have been predicted with the models fitted excluding them. Both models have been proved stable. They have similar predictions, but metrics show the better model.

Metric	SARIMA(2, 1, 0)(4, 1, 0)$_{12}$	SARIMA(0, 1, 1)(4, 1, 0)$_{12}$
RSME	0.810	0.804
MAE	0.665	0.660
RMSPE	0.035	0.346
MAPE	0.028	0.028
Mean conf. int. amplitude	5.076	4.755

Table 3 Prediction metrics of the outlier-adjusted models with calendar effects treatment

We can see how the SARIMA(0, 1, 1)(4, 1, 0)$_{12}$ has better prediction metrics, thus we choose this model for predicting over 2019.

In the following graphic, we can compare the predictions for 2019 of the original series models and the model for the outlier adjusted series with calendar effects treatment.

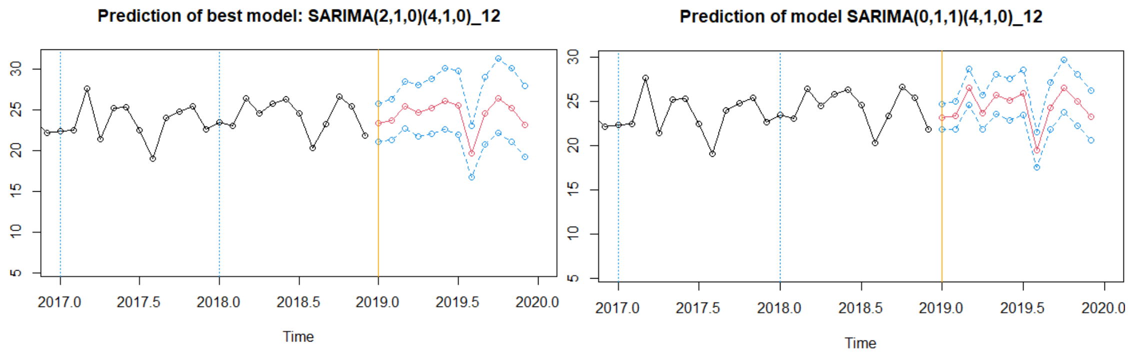

Figure 18 Predictions with the original model (left) and the outlier adjusted model (right) for 2019

We can see how the model adjusted for outliers gives more accurate predictions with a much thinner confidence interval. In particular, this are the confidence interval mean amplitude for both predictions:

- Mean confidence interval amplitude: 4.572
- Mean confidence interval amplitude: 7.124

Additionally, we see how the original model doesn't predict much variation at the first half of the year while the final model predicts higher and lower values for this period. Both models seem predict a year with average value similar to the past years. No significant increase or decrease of the exports is shown on the graphics.

5. Conclusions

To start extracting conclusions of the data we should first compare and decide the best model created so far. In order to do so, we need to compare fitting and prediction metrics for the data. We can use the following table:

	Par.	σ^2	AIC	BIC	RMSE	MAE	RMSPE	MAPE	CI
SARIMA	6	0.003	-673.88	-649.90	0.917	0.699	0.039	0.029	7.337
SARIMA+EC.	8	0.002	-786.53	-755.70	0.834	0.674	0.036	0.028	7.008
SARIMA+EC +OutTreat.	14	0.001	-892.45	-840.69	0.804	0.660	0.035	0.028	4.755

Table 4 Comparison of the fitting and prediction metrics of the 3 chosen models

Analysing the table, we can clearly select the SARMIA model with calendar effects and outlier treatment as the best. The calendar effects have a noticeable implication in the model fitting, since both AIC and BIC improve significantly in comparison to the original model. The prediction metrics do also improve, although they are not reduced by a great amount. On top of that, the validation of the model is the best of the three, ensuring independence and normality of residuals.

Outlier treatment has also improved considerably the fitting measures of in comparison to de SARIMA model with calendar effects treatment. In this case, the prediction metrics are much better after outlier treatment because error metrics have slightly decreased, but confidence interval mean amplitude is much lower.

In conclusion, both calendar effect treatment and outlier treatment are significant improvements to the model fitting and predictions and should be use while predicting over the future in this time series.

Regarding the future evolution of the exports in the next year, we have seen in Figure 14 that the prediction of the best model (calendar and outlier treatment model) has predicted a 2019 where Spain's exports will follow approximately last year mean value. Furthermore, a high peak of exports is predicted in March, and as usual a low amount of exports is predicted for the month of August. Nevertheless, this low peak should not be significantly worse than last year's one, and exports should increase until October and then slightly decrease until December.

6. References

1. Sede electrónica del Ministerio de Industria y Turismo. Web with Badase content. https://sede.serviciosmin.gob.es/es-ES/datosabiertos/catalogo/badase